

1. Complete the statements.

Patty paper is similar to a

- ☐ line.
- ☐ point.
- ☒ plane.

Drawing a figure on patty

paper and then turning the patty paper represents a

- ☐ reflection
- ☒ rotation
- ☐ translation

of the figure. Drawing a figure on patty paper and then moving the patty

paper to the right and up represents a

- ☐ reflection.
- ☐ rotation.
- ☒ translation.

Drawing a

figure on patty paper and then flipping the patty paper over represents a

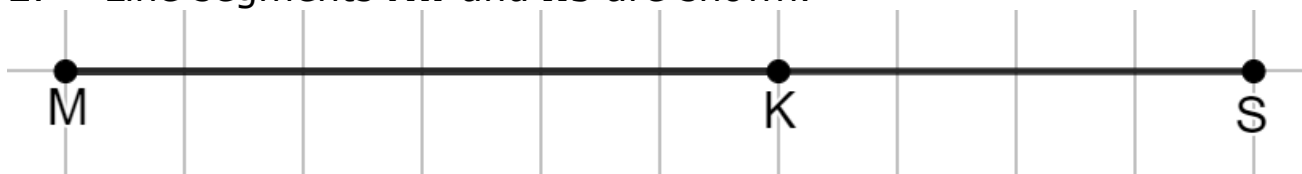
- ☒ reflection.
- ☐ rotation.
- ☐ translation.

Reflections, rotations, and translations are known

as

- ☐ mappings.
- ☒ rigid transformations.
- ☐ non-rigid transformations.

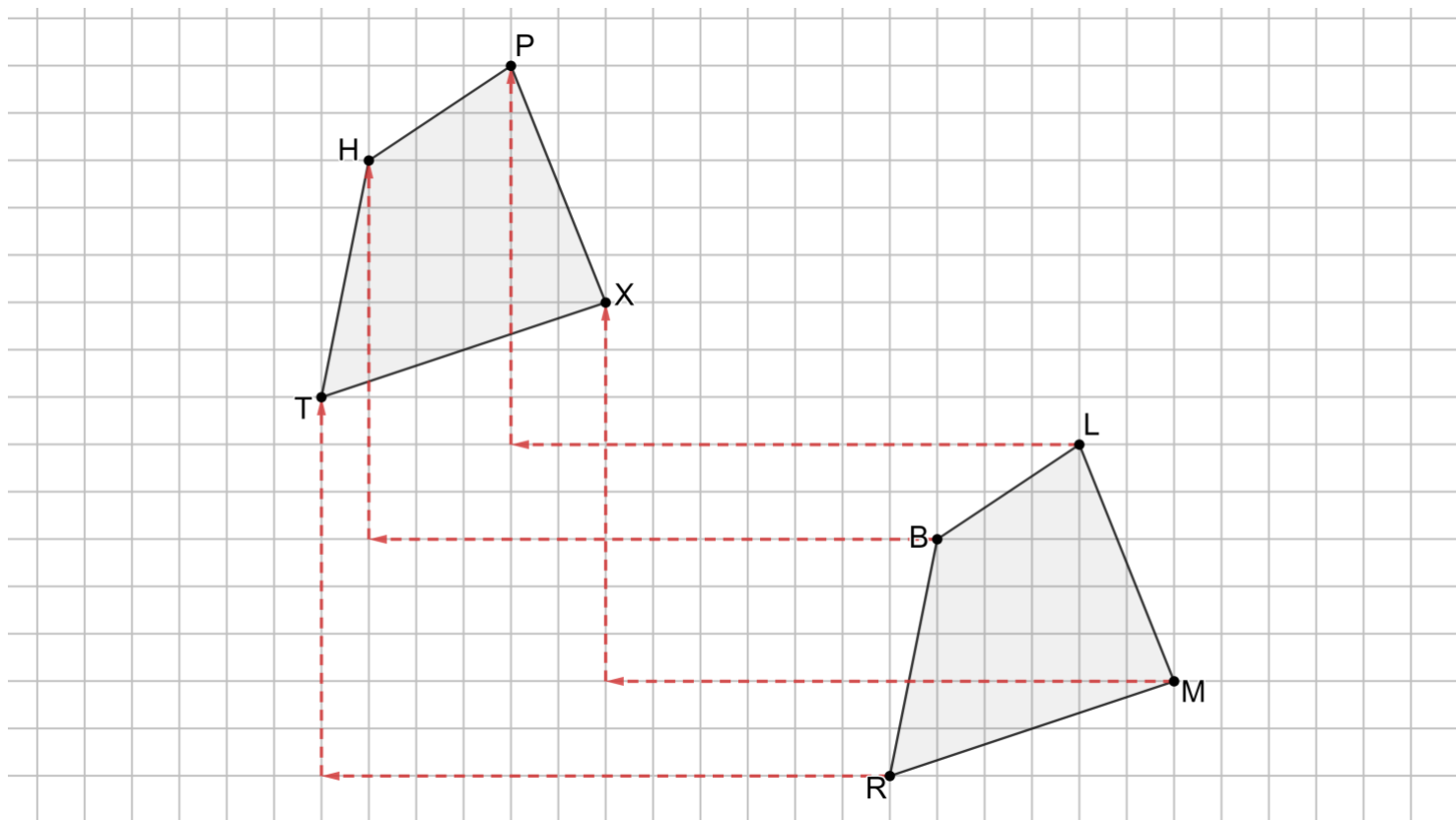
2. Line segments MK and KS are shown.



What is the scale factor can be used to dilate $\overline{MS}$ to create $\overline{MK}$?

$\frac{6}{10}$

3. Quadrilaterals $LMRB$ and $PXTH$ are shown. Draw the mapping of each vertex of $LMRB$ for the transformation that was used to create $PXTH$.



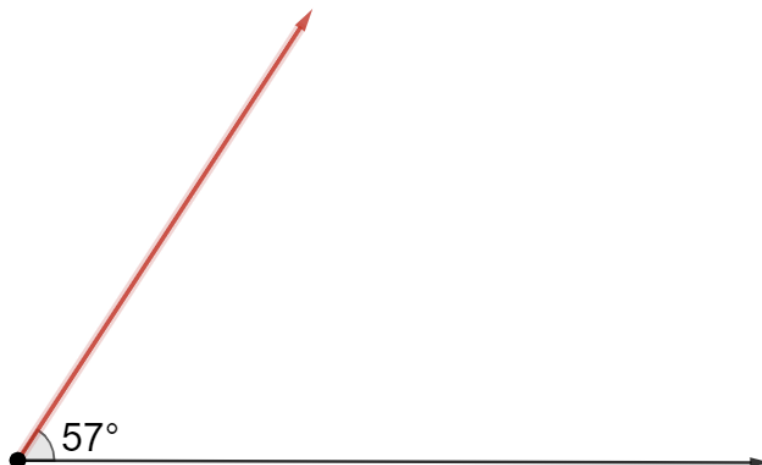
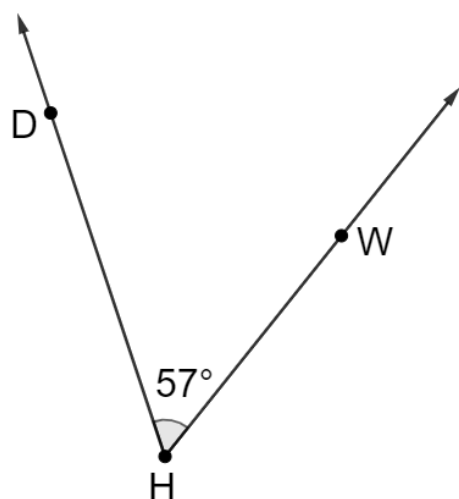
4. Points T , M , and B are collinear. Point T is between M and B .
 $MB = 12x + 44$, $BT = 11x - 1$, and $MT = 8x - 11$.
- a. Find the value of x . Show your work.

$x =$	8
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- b. What is the length of $\overline{MT}$?

$MT =$	53
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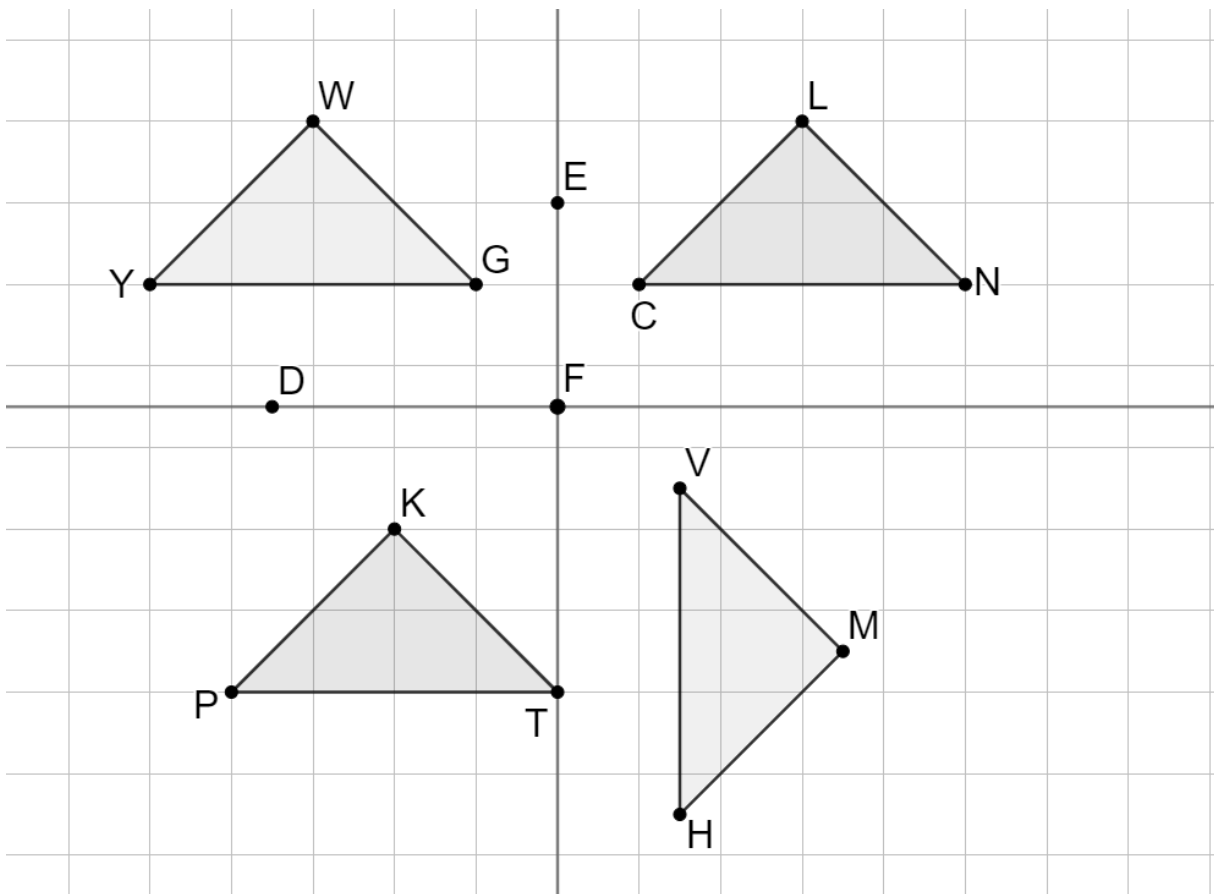
5. Use the given ray to construct a copy $\angle DHW$.



Complete the table for each description.

	Description	Defined or Undefined?	Term
6.	Lines that lie in the same plane and do not intersect.	☐ A defined term ☐ An undefined term	☐ parallel lines ☐ perpendicular lines ☐ skew lines ☐ parallel planes ☐ perpendicular planes
7.	An object that has infinite length, infinite width, and zero height.	☐ A defined term ☐ An undefined term	☐ line ☐ plane ☐ skew planes ☐ straight angle ☐ perpendicular lines

8. Four triangles are shown.



- a. Describe the single transformation of $\triangle LNC$ that was used to create $\triangle WYG$.

Reflection across line EF .

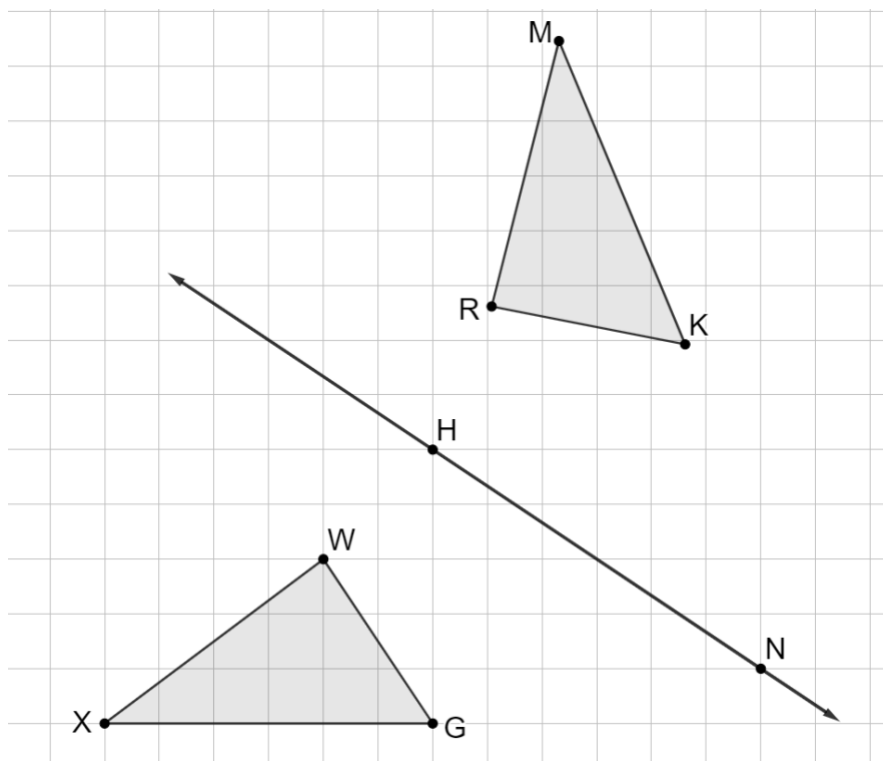
- b. Describe the single transformation of $\triangle LNC$ that was used to create $\triangle KTP$.

Translate left 5 down 5

- c. Describe the single transformation of $\triangle LNC$ that was used to create $\triangle MHV$.

Rotation 90° clockwise about point F

9. Triangles XGW and MKR are shown where ΔMKR is a transformation of ΔXGW .



- a. Complete the table to describe the transformation that was used to create ΔMKR .

Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☒ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is <u>HN</u> . ☐ Down ☐ Left ☐ Right ☐ Up	

Complete each statement.

b.

$$\overline{XW} \cong \overline{MR}$$

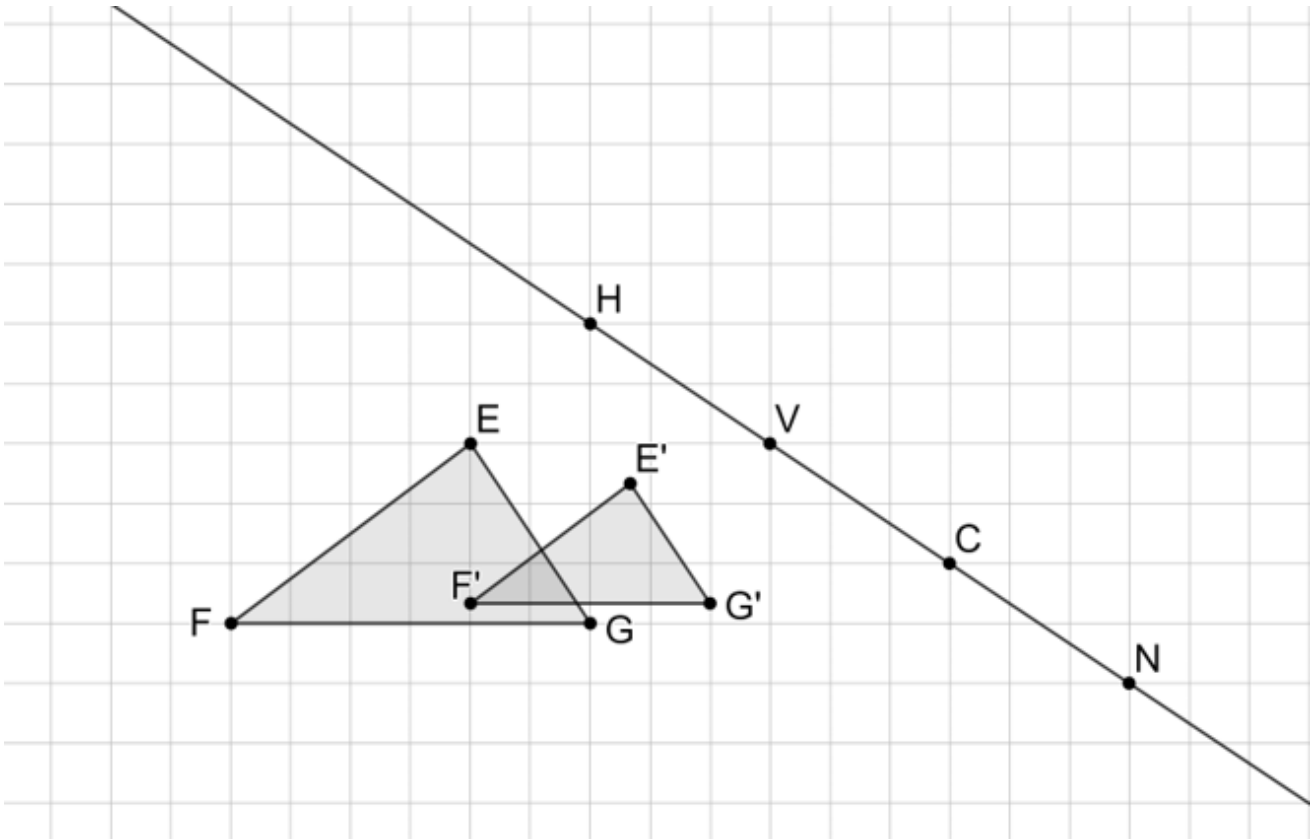
c.

$$\angle GWX \cong \angle MRK$$

d.

$$\overline{WG} \cong \overline{KR}$$

10. Triangles EFG and $E'F'G'$ are shown where $\Delta E'F'G$ is a transformation of ΔEFG .



Complete the table to describe the transformation that was used to create $\Delta E'F'G'$.

Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☒ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is C. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	$\frac{2}{3}$

11. $\overline{HD}$ is the perpendicular bisector of $\overline{KY}$ such that point D lies on $\overline{KY}$.
 $DK = 7y + 17$, $HY = 4y + 33$, $DY = 10y - 22$, and $HK = 6y + 7$.

a. Find the value of y .

$y =$	13
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b. What is the length of $\overline{KY}$?

$KY =$	216
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12. $\overrightarrow{KR}$ bisects $\angle PKN$ such that $m\angle PKR = 9x + 2$ and $m\angle NKR = 4x + 19.5$.

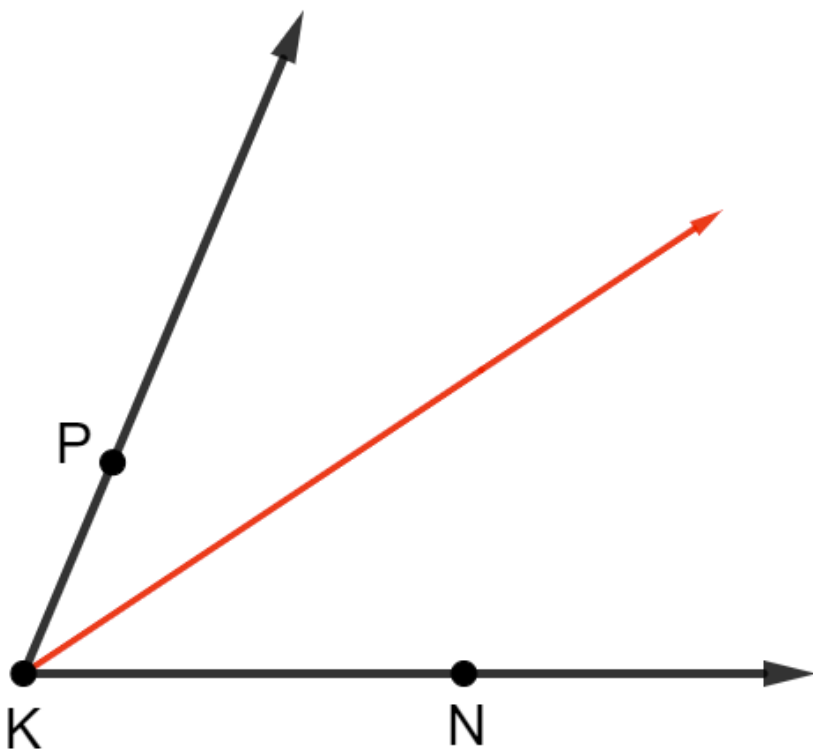
a. Find the value of x .

$x =$	3.5
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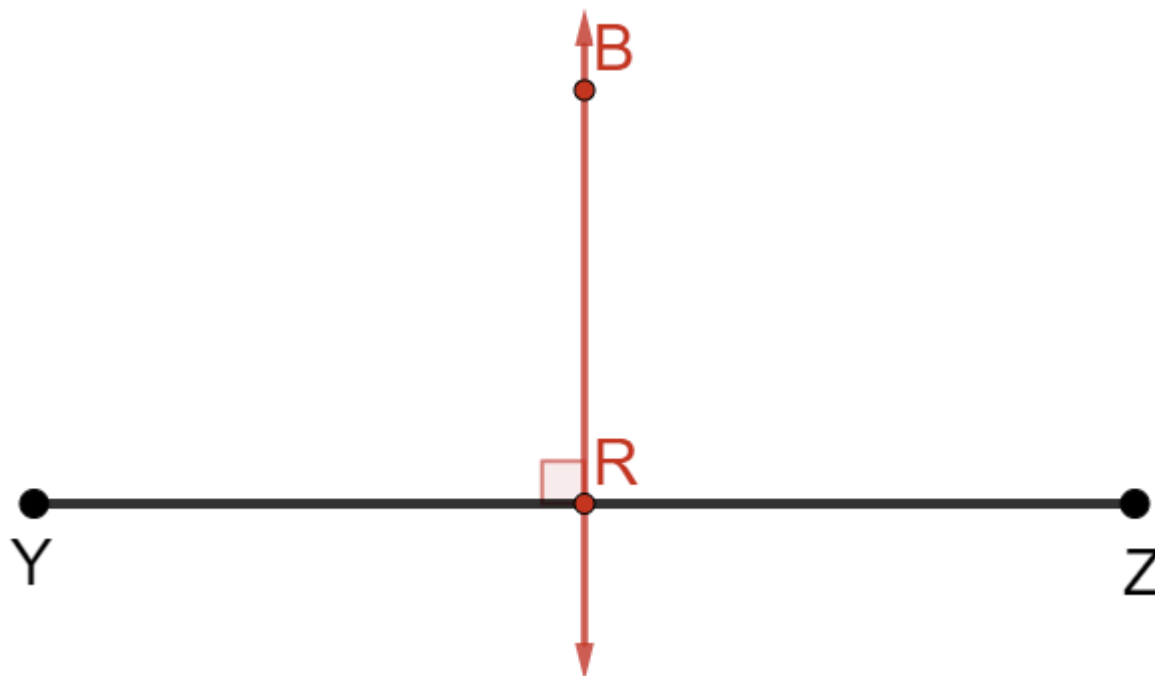
b. Determine $m\angle PKN$.

$m\angle PKN =$	67°
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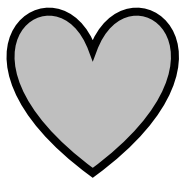
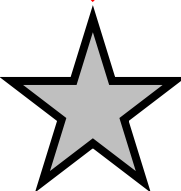
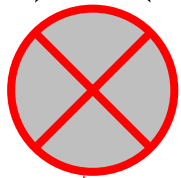
13. Construct the angle bisector of $\angle PKN$.



14. Construct the perpendicular bisector of $\overline{YZ}$.



15. Select all of the shapes whose order of rotational symmetry is 4.

☐☐☐☐☐